Week 10 Workshop

FIT5230 Malicious AI

AI Bias & Safety, Fairness, Explainable AI

MONASH University

Bias in AI

Generative AI Bias & Safety

Understanding Bias in Generative AI

- How biases in generative AI systems happen?
  - Through training data, model design, and deployment.

- AI chatbots like ChatGPT and Microsoft Copilot are trained using vast amounts of publicly accessible online text.
  - Do they have biases? – Yes! They would have the same biases as those sources and the society that produced them.

Type of Biases in AI

1. Cognitive and societal bias
    - Humans are biased.
    - This doesn't mean intentionally discriminatory.
    - It means humans make judgments based on:
        Experience
        Culture
        Beliefs
        Values

Researchers may collect data mainly from the United States because it is easier to obtain.
As a result, AI learns mostly American perspectives.

Type of Biases in AI

2. Cultural biases – prejudices and stereotypes
    - Different cultures think differently.

A Western dataset may define "successful leadership" differently from an Asian dataset.

AI learns those assumptions.

Type of Biases in AI

3. Training data bias
   - Occurs when datasets are not representative.

   Example:

   Face recognition dataset:

   80% white faces
   20% minority faces

   The model has far more learning opportunities for white faces.

   As a result, higher recognition accuracy for white individuals.

Type of Biases in AI

4. Algorithmic bias
   - This bias occurs during model construction.
   - Even if data looks fair, algorithms may still create unfair outcomes.

   Example:
   Suppose a bank AI uses:
   Income
   Address
   Education

   Income may indirectly correlate with demographics.
   The AI could unfairly favor certain groups.

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Race not used, postcode used

But postcode strongly correlates with race.

Mitigating Bias in Generative AI

Diverse data collection

- Use more representative datasets.

Instead of:

Mostly male
Mostly Western

Use broader representation.

Why?

Mitigating Bias in Generative AI

1. Diverse data collection
   - Use more representative datasets.

Instead of:

Mostly male
Mostly Western

Use broader representation.

Why?

Because AI cannot learn patterns it never sees.

Garbage in → garbage out.

Biased data in → biased model out.

Mitigating Bias in Generative AI

2. Algorithmic auditing
- Regularly inspect outputs.

Questions:

Is one group rejected more often?
Is one group misclassified more often?

Purpose is to find hidden discrimination.

Tools:
- AI Fairness 360
- Themis-ML
- What-If Tool

Mitigating Bias in Generative AI

3. Algorithmic Fairness Techniques
- Methods built directly into training.

Example:
- Adversarial debiasing
- Fairness constraints

4. Transparency and explainability in AI algorithms
- If users understand decisions:
  - Bias becomes easier to detect.
  - Organisations become more accountable.

Mitigating Bias in Generative AI

5. Regulatory and Ethical Frameworks
- Set standards for fairness, accountability, and transparency

6. Human-in-the-loop Systems
- Humans review important decisions.
  Example:

  AI suggests loan rejection.

  Human officer reviews before approval.

  This reduces fully automated discrimination.

Mitigating Strategies: Adversarial Debiasing

Core Idea: Two neural networks compete

Main Network
Learns the primary task. E.g., Classifying faces

Adversary Network
Attempts to predict protected attributes, e.g., gender, race

Training Goal

- Force the main model to remove information related to protected attributes.
- Eventually, the adversary should fail.
- If the adversary cannot predict gender:
- Gender information has largely disappeared from learned features.

Adversarial Debiasing: AGENDA

Towards Gender-Neutral Face Descriptors for Mitigating Bias in Face Recognition

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Abstract

State-of-the-art deep networks implicitly encode gender information while being trained for face recognition. Gender is often viewed as an important attribute with respect to identifying faces. However, the implicit encoding of gender information in face descriptors has two major issues: (a.) It makes the descriptors susceptible to privacy leakage, i.e. a malicious agent can be trained to predict the face gender from such descriptors. (b.) It appears to contribute to gender bias in face recognition, i.e. we find a significant difference in the recognition accuracy of DCNNs on male and female faces. Therefore, we present a novel 'Adversarial Gender De-biasing algorithm (AGENDA)' to reduce the gender information present in face descriptors obtained from previously trained face recognition networks. We show that AGENDA significantly reduces gender predictability of face descriptors. Consequently, we are also able to reduce gender bias in face verification while maintaining reasonable recognition performance.

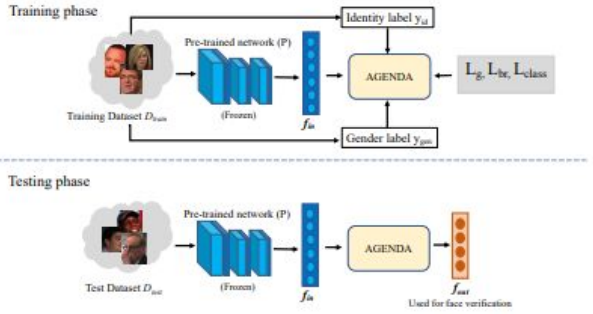


Figure 1: We present a framework called AGENDA, which learns to reduce gender information from face descriptors f_{in}. After training, we can transform the face descriptors using AGENDA and generate their gender agnostic representations f_{out}, which can then be used for face recognition/verification.

Adversarial Debiasing: AGENDA

Recognition model shows bias towards particular race,gender or ethnicity.

- Gender is often found as most important facial feature while classifying the images → big impact in the performance
- Gender bias
- Achieving more accuracy for a particular gender for same model and training data

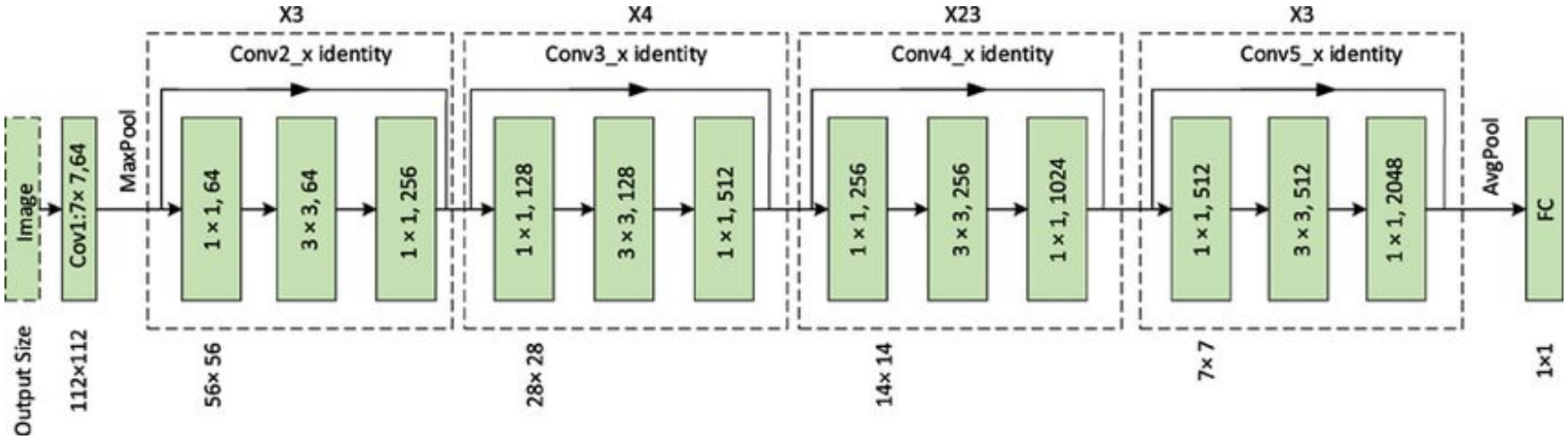
Identifying the bias:

- Two different datasets Network A and Network B
- Network A : There are 59,563 males and 22,499 females
- Network B : There are 39,712 males and 18,308 females

Adversarial Debiasing: AGENDA

Identifying the bias (cont.)

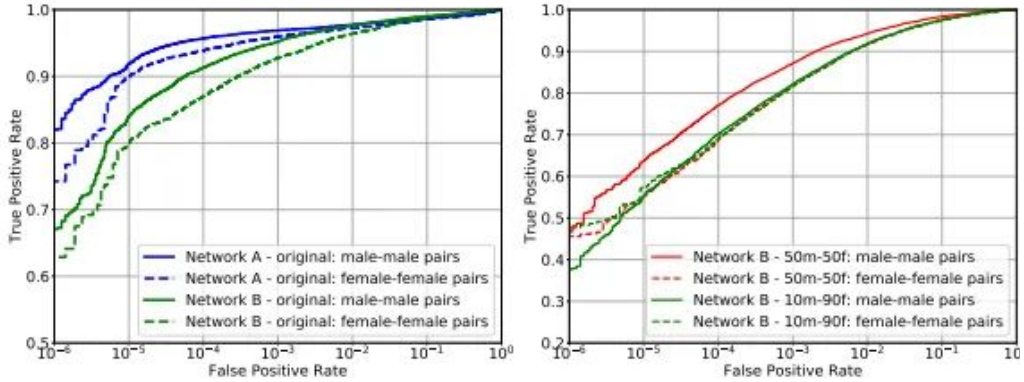
- Train Resnet-101 on both datasets
- Compare its performance before and after balancing the dataset



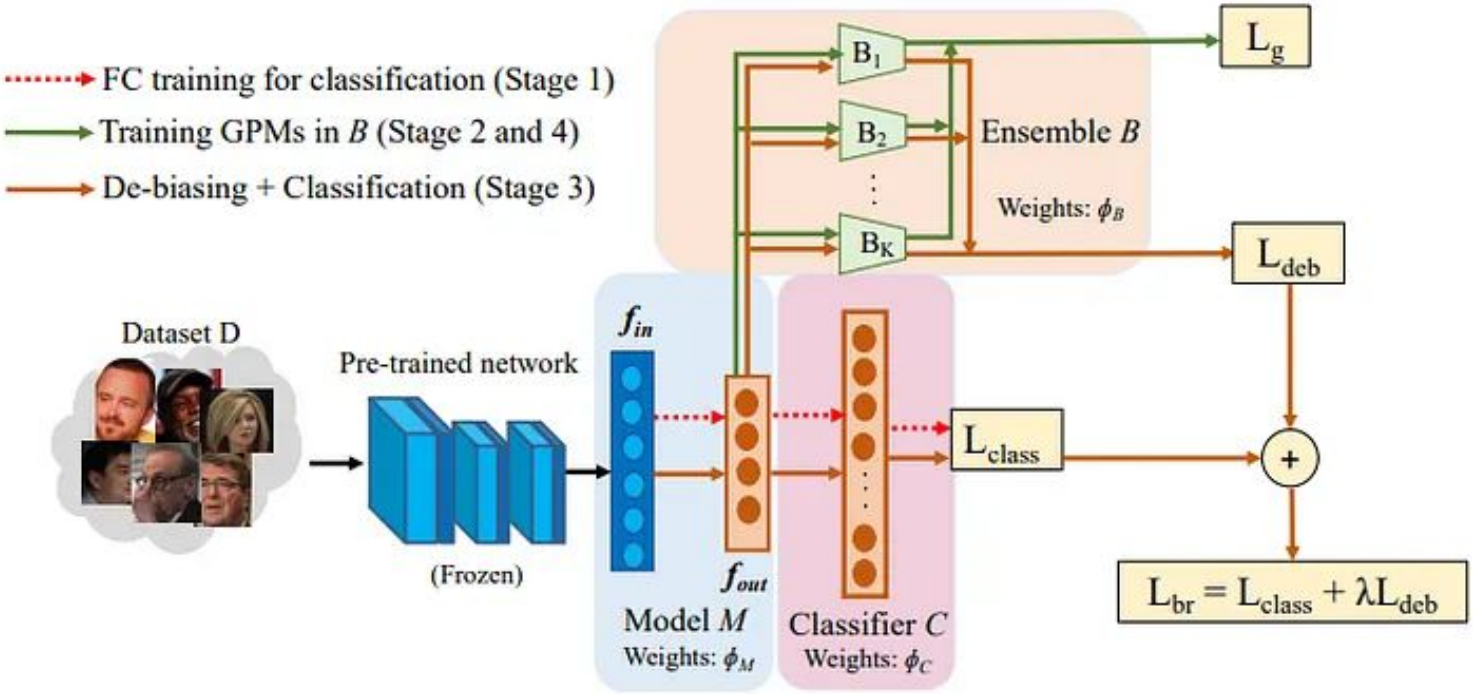
Adversarial Debiasing: AGENDA

Compare original dataset's male-male and female female pairs.

- Extract training example in different ratios of female and male labels from Network B and compare the results
  - 50% male-50% female
  - 10% male -90% female



Adversarial Debiasing: AGENDA



Adversarial Debiasing: AGENDA

Generator model M:

- Takes in dataset f_in from a pre-trained network P, and generates a lower dimensional descriptor f_out
- M consists of a single linear layer with 256 units, followed by a PReLU layer. The weights of M are denoted as ϕM

Classifier C:

- A classifier that takes in the output of M(fout) and generates a prediction vector for identity classification. The weights of C are denoted as ϕC

Ensemble of discriminators B:

- An ensemble of K gender prediction models represented as B1, B2 . . . BK

Adversarial Debiasing: AGENDA

Ensemble of discriminators B (cont.):

- Each of the ensemble model consist of:
- A simple MLP network with an input layer size of 256 units
- A SELU activated linear layer with 128 units,
- A sigmoid activated output layer with 2 units
- A softmax layer

The weights of all the models in B is denoted as ϕB and weights of k-th model Bk as ϕBk .

Adversarial Debiasing: AGENDA

The total loss function defined for Model M:

$$L_{br}(\Phi_C , \Phi_M, \Phi_B) = L_{class}(\Phi_C , \Phi_M) + \lambda L_{deb}(\Phi_M, \Phi_B)$$

- L_{class} = loss of the classifier C
- L_{deb} = loss of ensemble B

Aim: Penalize M, with respect to the strongest ensemble model K

- Strongest ensemble model → strongest discriminator
- Update L_{deb} till the strongest ensemble model does not give 0.5 probability.
- The method works for both Network A and B.

Challenges in Resolving Bias

Complex and multifaceted data

- AI systems learn from the data they are trained on
- Identify and correct biases in the data
- Rethink how data is collected, curated, and utilized

Complex of modern AI systems

- Generative AI models are has millions or even billions of parameters
- Operate as black boxes → lack of explainability

No universal solution/framework

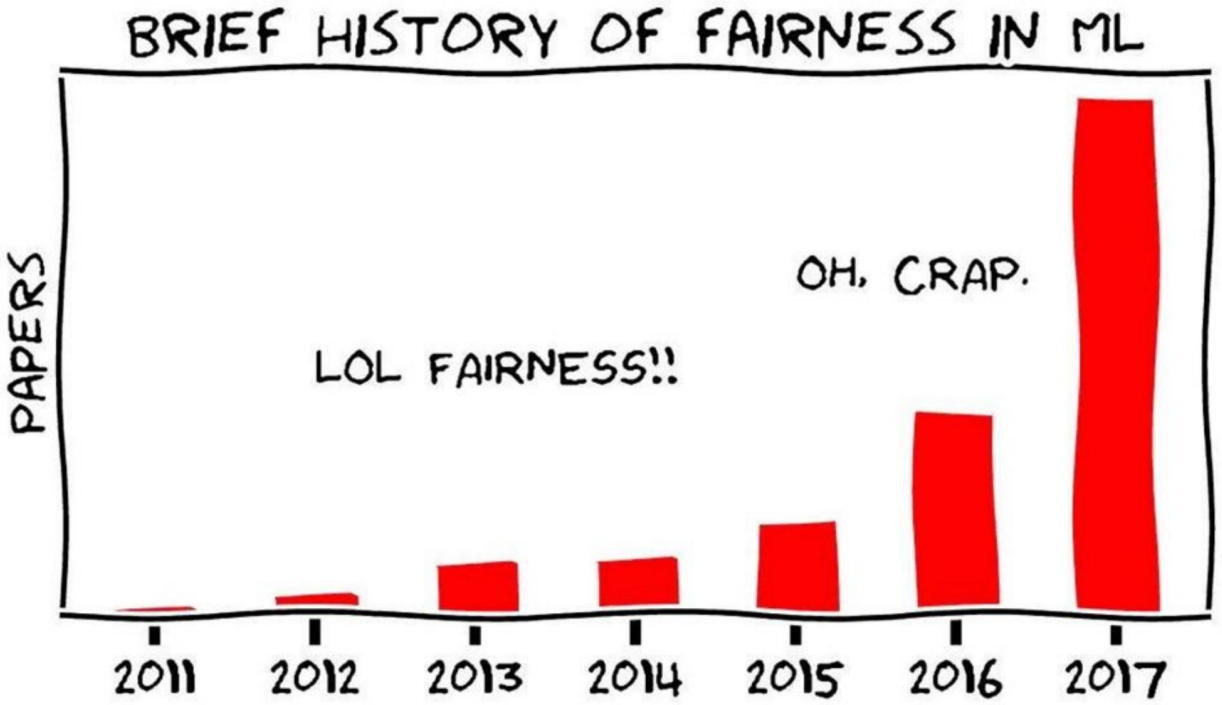
- Different types of bias, i.e., racial bias, gender bias, and cultural bias
- biases are often context-dependent

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Fairness

Fair Machine Learning

Fair Machine Learning



Fair Machine Learning - Challenges

1. **Defining fairness is hard**
   - Many competing definitions, often incompatible with each other
     - statistics-based, causal-reasoning-based, etc.
     - focus on group outcomes vs. individual outcomes
   - Requires contextual understanding

Name	Closest relative	Note
Statistical parity	Independence	Equivalent
Group fairness	Independence	Equivalent
Demographic parity	Independence	Equivalent
Conditional statistical parity	Independence	Relaxation
Equal opportunity	Separation	Relaxation
Equalized odds	Separation	Equivalent
Conditional procedure accuracy equality	Separation	Equivalent
Disparate mistreatment	Separation	Equivalent
Balance for positive class	Separation	Relaxation
Balance for negative class	Separation	Relaxation
Predictive equality	Separation	Relaxation
Conditional use accuracy equality	Sufficiency	Equivalence
Predictive parity	Sufficiency	Relaxation
Calibration	Sufficiency	Equivalence

Fairness definition - Example

demographic parity: “f(x) be unconditionally independent of sensitive attr. a.”

- if 100 female students and 100 male students apply to Monash University, **demographic parity** is achieved if the percentage of female students admitted is the **same** as the percentage of male students admitted, irrespective of whether one group is on average more qualified than the other.
- This metric focuses on equal outcomes.

Fairness definition - Example

equalized odds: “f(x) be independent of sensitive attr. a given y”

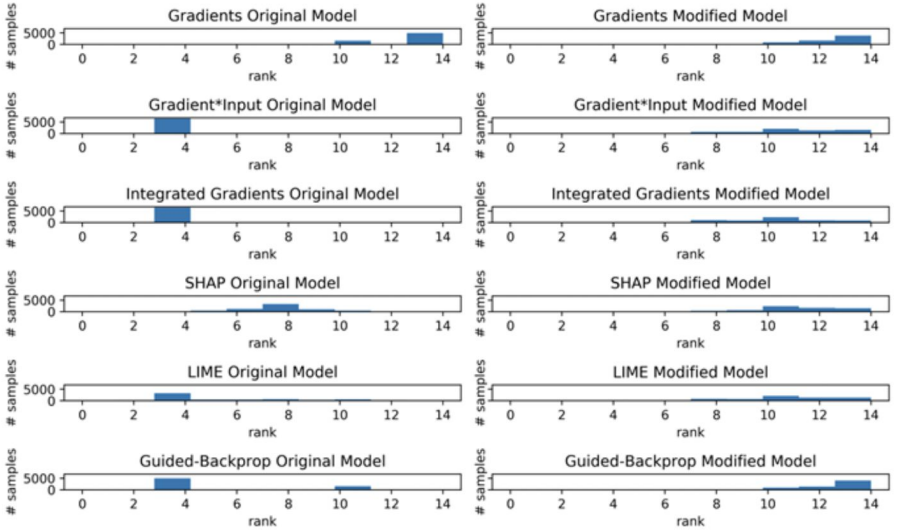
- if 100 female students and 100 male students apply to Harvard University, equalized odds is achieved if **qualified** female and male students both have the **same** chance of being admitted, and **unqualified** female and male students have the **same** chance of being rejected.
- This metric focuses on correctness and fairness simultaneously

Fair Machine Learning - Challenges

2. Explanation methods can be manipulated

Demonstrates that an explanation attack can easily mask a model's discriminatory use of a sensitive feature without hurting accuracy.

Importance ranking histograms for gender as the sensitive feature on the adult test set of the original (left) and modified (right) models.



Reference: You shouldn't trust me: Learning models which conceal unfairness from multiple explanation methods, Dimanov, ECAI, 2020

Fair ML - Proposed Solution

- A unified approach that works for many group-fairness criteria
  - demographic parity, equalised odds, conditional demographic parity
  - for each definition, choose Shapley value functions that attribute overall fairness to individual features.
- Cannot hide unfairness by manipulating explanations
  - **Fairness Shapley values** collectively must sum to the chosen fairness metric

Fairness Shapley Value

Imagine a loan approval model uses:

- Income
- Age
- Education
- Gender

and predicts:

Loan Approved

You want to know:

Which features contributed the most to this decision?

Fairness Shapley Value

Shapley Values come from game theory.

In game theory:

- Players work together to earn a reward.
- Shapley Value measures each player's contribution to the final reward.

In ML:

- Features are the players.
- Prediction is the reward.

Fairness Shapley Value

Suppose a university admission system uses: exam score, interview score and gender.

The model rejects a candidate.

Shapley analysis shows:

Feature	Contribution
Exam score	+0.35
Interview	+0.20
Gender	-0.40

Fairness Shapley Value

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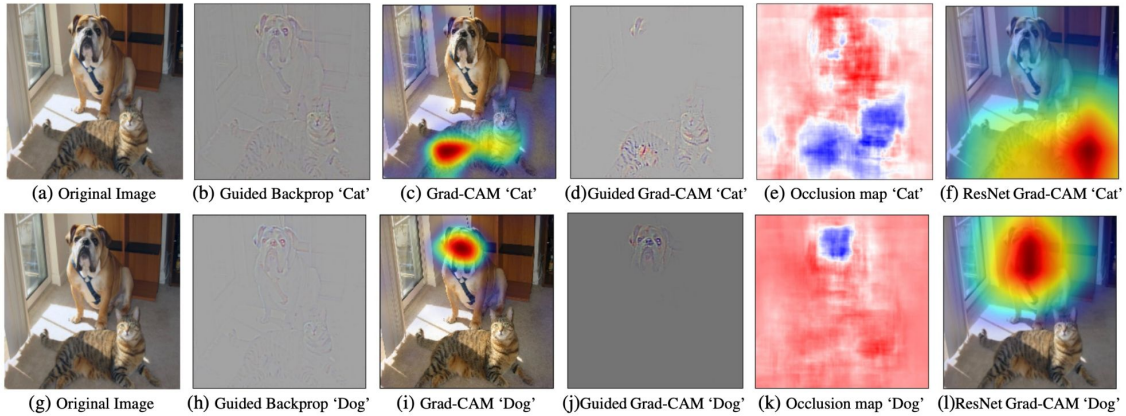
This is suspicious

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Explainable AI (XAI)
Computer Vision

Gradient weighted Class Activation Mapping (Grad-CAM)

- Weight the 2D activations by the average gradient
- Uses gradients of a particular target that flows through the convolutional network to localize and highlight regions of the target in the image



(a) Original Image (b) Guided Backprop 'Cat' (c) Grad-CAM 'Cat' (d)Guided Grad-CAM 'Cat' (e) Occlusion map 'Cat' (f) ResNet Grad-CAM 'Cat'

(g) Original Image (h) Guided Backprop 'Dog' (i) Grad-CAM 'Dog' (j)Guided Grad-CAM 'Dog' (k) Occlusion map 'Dog' (l)ResNet Grad-CAM 'Dog'

Fig. 1: (a) Original image with a cat and a dog. (b-f) Support for the cat category according to various visualizations for VGG-16 and ResNet. (b) Guided Backpropagation [53]: highlights all contributing features. (c, f) Grad-CAM (Ours): localizes class-discriminative regions, (d) Combining (b) and (c) gives Guided Grad-CAM, which gives high-resolution class-discriminative visualizations. Interestingly, the localizations achieved by our Grad-CAM technique, (c) are very similar to results from occlusion sensitivity (e), while being orders of magnitude cheaper to compute. (f, l) are Grad-CAM visualizations for ResNet-18 layer. Note that in (c, f, i, l), red regions corresponds to high score for class, while in (e, k), blue corresponds to evidence for the class. Figure best viewed in color.

Reference: Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization

Gradient weighted Class Activation Mapping (Grad-CAM)

Grad-CAM Architecture

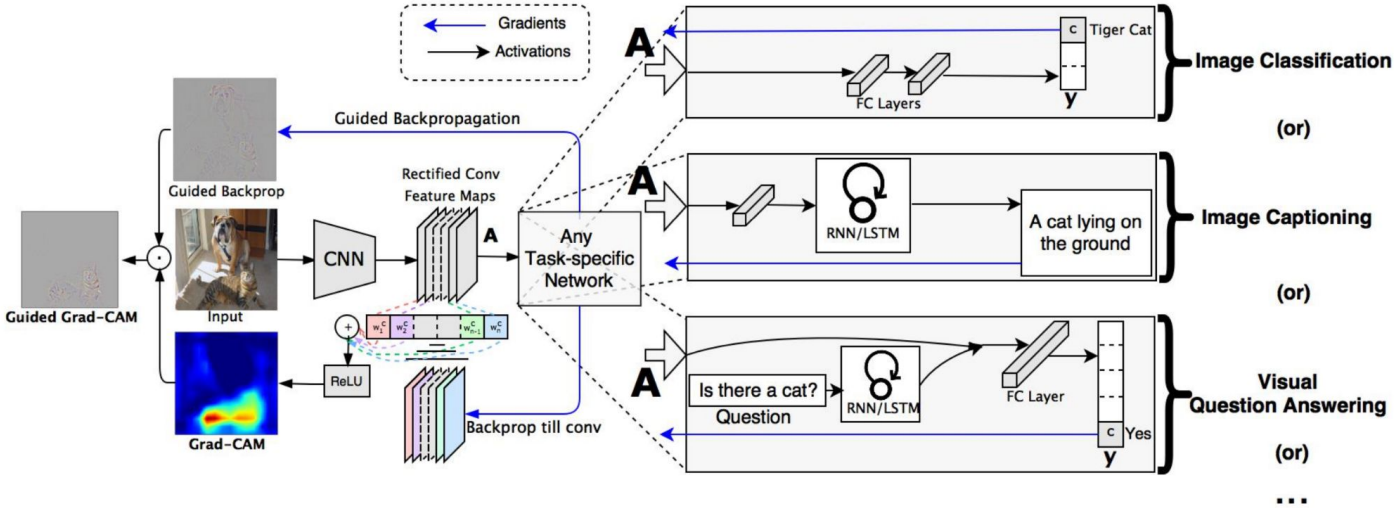


Fig. 2: Grad-CAM overview: Given an image and a class of interest (*e.g.*, 'tiger cat' or any other type of differentiable output) as input, we forward propagate the image through the CNN part of the model and then through task-specific computations to obtain a raw score for the category. The gradients are set to zero for all classes except the desired class (tiger cat), which is set to 1. This signal is then backpropagated to the rectified convolutional feature maps of interest, which we combine to compute the coarse Grad-CAM localization (blue heatmap) which represents where the model has to look to make the particular decision. Finally, we pointwise multiply the heatmap with guided backpropagation to get Guided Grad-CAM visualizations which are both high-resolution and concept-specific.

Reference: Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization

Gradient weighted Class Activation Mapping (Grad-CAM)

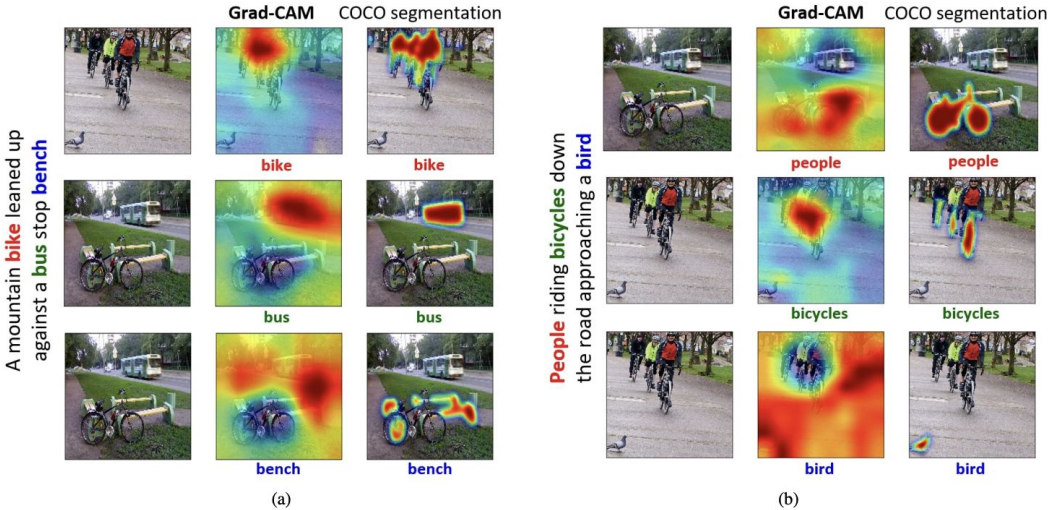


Fig. 11: Qualitative Results for our word-level captioning experiments: (a) Given the image on the left and the caption, we visualize Grad-CAM maps for the visual words "bike", "bench" and "bus". Note how well the Grad-CAM maps correlate with the COCO segmentation maps on the right column. (b) shows a similar example where we visualize Grad-CAM maps for the visual words "people", "bicycle" and "bird".

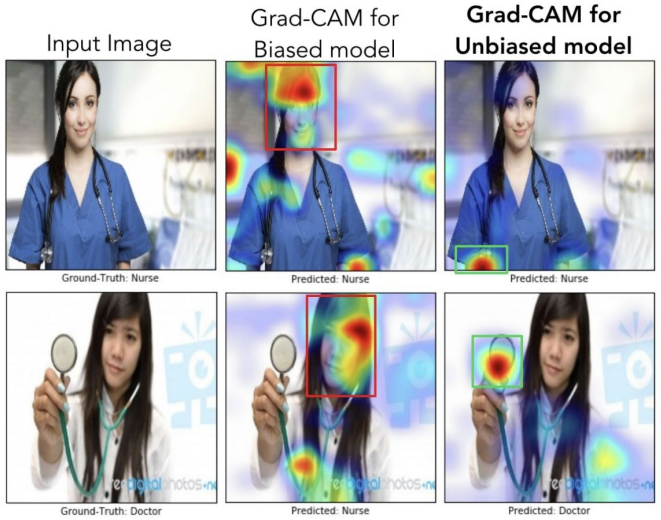


Fig. 8: In the first row, we can see that even though both models made the right decision, the biased model (model1) was looking at the face of the person to decide if the person was a nurse, whereas the unbiased model was looking at the short sleeves to make the decision. For the example image in the second row, the biased model made the wrong prediction (misclassifying a doctor as a nurse) by looking at the face and the hairstyle, whereas the unbiased model made the right prediction looking at the white coat, and the stethoscope.

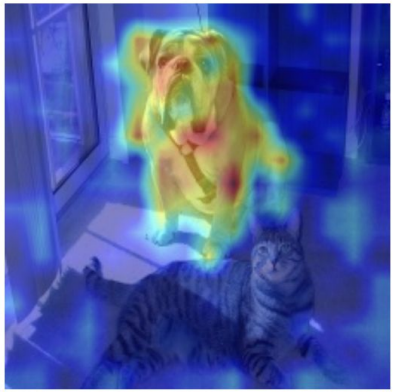
Reference: Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization

Gradient weighted Class Activation Mapping (Grad-CAM)

Explaining similarity to other images / embeddings



CLIP

Explaining the text prompt "a dog"	Explaining the text prompt "a cat"
	

Reference: Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization

AI Bias

Which of the following statements about AI bias are correct?

I. AI bias can result from historical biases present in the training data.
II. AI bias can be mitigated by using techniques like re-sampling and re-weighting the training data.
III. AI bias is only a concern in image recognition systems.
IV. AI bias can occur due to biased feature selection.
V. AI bias is always intentional.

Share your answer here: https://shorturl.at/ZqE07

AI Bias

Which of the following strategies can help reduce AI bias?

I. Ensuring that only technical experts are involved in the development of AI systems.
II. Incorporating fairness constraints during the model training process.
III. Regularly auditing AI systems for biased outcomes.
IV. Ignoring the issue of bias to focus on improving model accuracy.
V. Using diverse and representative datasets for training.

Share your answer here: https://shorturl.at/ZqE07

AI Bias

Which of the following are correct ways to address AI bias?

I. Engaging diverse teams in the development and testing of AI systems.
II. Using explainable AI techniques to understand and mitigate bias.
III. Implementing continuous monitoring and evaluation of AI systems.
IV. Relying solely on automated tools to detect and correct bias.
V. Assuming that AI systems are inherently unbiased and require no further scrutiny.

Share your answer here: https://shorturl.at/ZqE07

AI Bias & Fairness

- Radhika had been out of the workforce for a few months, due to her pregnancy and taking care of her newborn. She had now returned to the job market, eager to get back to work and resume her career. However, her job search was not going as planned.
- She had applied to several positions but had not received any calls for an interview. Radhika couldn't understand why, as she had a strong resume, relevant experience, and a solid education.

Questions:

- Identify the possible bias.
- Provide 2 strategies to mitigate bias in the system.

Share your answer here: https://shorturl.at/ZqE07

AI Bias & Fairness

- SecureBank, that uses a machine learning model to detect fraudulent transactions. The model analyzes various features of each transaction, such as the amount, location, time, and the user's transaction history, to determine the likelihood of fraud.
- A group of attackers, aiming to exploit this system, creates adversarial examples. These are slightly modified versions of legitimate transactions designed to deceive the model. The attackers use techniques like adding small, imperceptible noise to the transaction data, which causes the model to misclassify fraudulent transactions as legitimate.

Share your answer here: https://shorturl.at/ZqE07